Quantum Mechanics II — Exercise Sheet 3

Boundary conditions and momentum

Variant: Questions only

Exercise 1 (A constant vector under twisted momentum). Let $\psi(x) = L^{-1/2}$ on $(0, L)$. For the self-adjoint momentum P_θ, with $0 < \theta < 2\pi$, find every momentum probability. (i) Decide whether $\psi \in D(P_\theta)$ and whether the second momentum moment is finite. (ii) Treat $\theta = 0$ separately.

Exercise 2 (Neumann energy is a different observable). Consider $H_N = -\hbar^2 \mathrm{d}^2/(2m\mathrm{d}x^2)$ with Neumann conditions $\psi'(0) = \psi'(L) = 0$. You may take as given that this operator is self-adjoint and that the cosine basis is complete. (i) Find its normalized eigenfunctions and energies, including the lowest one. (ii) Compare its ground state with the Dirichlet box ground state.

Exercise 3 (A phase changes the boundary condition). Let $(G_\theta f)(x) = e^{\mathrm{i}\theta x/L} f(x)$ on $L^2(0, L)$. (i) Show that $G_\theta D(P_0) = D(P_\theta)$ and compute $G_\theta^{-1} P_\theta G_\theta$ on $D(P_0)$. (ii) Explain why this is not an equality of P_0 and P_θ on the same domain.

Exercise 4 (Zero current does not choose a momentum). Let ψ be a real, smooth, normalized function with compact support inside $(0, L)$. (i) Show that it belongs to every $D(P_\theta)$ and that its probability current is zero. (ii) Explain why the momentum distributions for two distinct twists need not be the same. State a feature of their possible outcomes that distinguishes the observables.